

# Joseph Siracusa

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## EDUCATION

### Northeastern University

*Bachelor of Science in Computer Science*

Expected December 2027

*Boston, MA*

**Coursework:** Algorithms & Data Structures, Computer Systems, Computer Architecture, Object Oriented Design, Foundations of Cybersecurity, Fundamentals of CS 1 & 2, Discrete Structures

## EXPERIENCE

### Northeastern University

*Software Engineering Research Assistant*

Jun. 2026 – Present

*Boston, MA*

- Contributing to a research pipeline to trace and rebuild Wasm binaries in NPM packages, helping close security gaps
- Wrote a Rust CLI to normalize Wasm modules into fingerprints mapped to upstream versions, flagging unpatched libraries
- Swept libraries across compilers and optimization levels, isolating invariants across toolchains to inform version matching

### NExT Consulting

*Software Engineer Intern*

Sep. 2025 – Dec. 2025

*Boston, MA*

- Built **12 AI-native prototypes** using multi-agent workflows and prompt engineering to guide Northeastern's AI strategy
- Evaluated **50 developer tools** such as Arize Phoenix, LangChain and CrewAI to find the best-fit solution for the university
- Automated CI/CD with GitHub Actions, Docker and Render, eliminating integration bugs before production

### Wyzant

*Software Engineer Tutor*

Jun. 2023 – Sep. 2025

*Remote*

- Delivered SWE tutoring to students from high school to graduate level, tailoring sessions to academic and career goals
- Tutored 40+ students in DSA, test-driven development and DB design achieving **100% satisfaction** over **300+ hours**
- Designed interactive exercises using Replit & GitHub for students to apply programming concepts with hands-on practice

### Northeastern University

*Computer Science Teaching Assistant*

Jan. 2025 – Apr. 2025

*Boston, MA*

- Provided instruction to **170+ students** on data structures, algorithms, and object-oriented programming in Java
- Mentored students on data structure selection and redundant computation over **10+ assignments** to optimize time/space
- Reinforced inheritance, polymorphism and encapsulation with **15+ code reviews**, improving student code structure

## PROJECTS

### Breathe Easy | Dart, Flutter, Google Map & Air Quality APIs, Docker, Git

1st Overall – CS Base Climate Hack

- Built a mobile app to compute optimal travel routes from real-time air quality data to help users avoid pollution hotspots
- Developed reactive Flutter/Dart UI for intuitive route visualization, helping users better prioritize their respiratory health
- Integrated Google Maps & Air Quality APIs, mapping route segments using the UAQI to compare exposure between routes

### Portal Card | Dart, Flutter, Google Wallet API, Git

3rd Overall – FinHacks 2025

- Created a demo mobile app for a hybrid credit-transit card concept to promote public transit adoption and sustainability
- Implemented the app in Flutter and Dart, managing state, UI and business logic to deliver a functional end-to-end demo
- Leveraged digital wallet integration via the Google Wallet API, enabling users to add Portal Cards to their wallets

### Math Monsters | HTML/CSS, JavaScript, Git

1st in Track – PawHacks 2024

- Constructed an interactive educational game to teach children foundational mathematics through customizable gameplay
- Utilized HTML, CSS, and JavaScript to implement game mechanics and provide an inviting UI for children

### City Story | JavaScript, React, Google Maps API, Java, Spring, MySQL, Git

- Produced a web app for tourists to discover, rate and travel to **30+ landmarks** with crowd-sourced recommendations
- Engineered a RESTful rating system with Spring Java & MySQL, allowing users to rate landmarks and guide future visitors

## TECHNICAL SKILLS

**Languages:** Java, Kotlin, Dart, Python, C, C#, HTML, CSS, JavaScript, Typescript, Go, SQL, Ocaml, x86-64 ASM, WASM

**Frameworks:** React, Node.js, JUnit, Spring, Ktor, Flask, FastAPI, LangChain, Flutter, .NET, GRPC

**Developer Tools:** Git, GitHub Actions, Docker, AWS, Linux, Redis, NGINX, Notion, Linear, Microsoft Office

**Concepts:** Data Engineering, System Design, Cloud Computing, OOP, SOLID principles, Unit Testing, DevOps, APIs, Operating Systems, Concurrency & Multithreading, Web Services, Prompt Engineering, Multi-Agent Systems